

Appendix:

A. Formula and Definition

1. Euler Expansion: $e^{j\pm x} = \cos x \pm j \sin x$.
2. Continuous-time Convolution: $x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$
3. Discrete-time Convolution: $x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$
4. Continuous-time Fourier Series: $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$, $a_k = \frac{1}{T} \int_T x(t) e^{-jk(2\pi/T)t} dt$
5. Discrete-time Fourier Series: $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$, $a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$
6. Continuous-time Fourier Transform: $x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$, $X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$
7. Discrete-time Fourier Transform: $x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$, $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$
8. CT Periodic Signal Fourier Transform: $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$, $X(j\omega) = 2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$

B. Properties of CTFT

Time Shifting	$x(t-t_0)$	$e^{-j\omega t_0} X(j\omega)$
Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega-\omega_0))$
Time Reversal	$x(-t)$	$X(-j\omega)$
Time Scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega-\theta))d\theta$
Differentiation in Time	$\frac{d}{dt} x(t)$	$j\omega X(j\omega)$
Integration	$\int_{-\infty}^t x(t)dt$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
Differentiation in frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$

C. Properties of DTFT

Time Shifting	$x[n-n_0]$	$e^{-jn\omega_0} X(e^{j\omega})$
Frequency Shifting	$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega-\omega_0)})$
Time Reversal	$x[-n]$	$X(e^{-j\omega})$
Time Expansion	$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = \text{multiple of } k \\ 0, & \text{if } n \neq \text{multiple of } k \end{cases}$	$X(e^{jk\omega})$
Multiplication	$x[n]y[n]$	$\frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$
Differencing in Time	$x[n] - x[n-1]$	$(1 - e^{-j\omega})X(e^{j\omega})$
Differentiation in Frequency	$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$

D. Fourier Transform

1. $\delta(t-t_0) \xrightarrow{\mathcal{F}} e^{-j\omega t_0}$ $u(t) \xrightarrow{\mathcal{F}} \frac{1}{j\omega} + \pi\delta(\omega)$ $e^{j\omega_0 t} \xrightarrow{\mathcal{F}} 2\pi\delta(\omega-\omega_0)$
2. $\cos \omega_0 t \xrightarrow{\mathcal{F}} \pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$ $\sin \omega_0 t \xrightarrow{\mathcal{F}} \frac{\pi}{j}[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)]$
3. $x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases} \xrightarrow{\mathcal{F}} \frac{2 \sin \omega T_1}{\omega} \quad \frac{\sin Wt}{\pi t} \xrightarrow{\mathcal{F}} X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$
4. $e^{-at}u(t), \text{Re}\{a\} > 0 \xrightarrow{\mathcal{F}} \frac{1}{a+j\omega}$; $te^{-at}u(t), \text{Re}\{a\} > 0 \xrightarrow{\mathcal{F}} \frac{1}{(a+j\omega)^2}$
5. $\frac{t^{n-1}}{(n-1)!}e^{-at}u(t), \text{Re}\{a\} > 0 \xrightarrow{\mathcal{F}} \frac{1}{(a+j\omega)^n}$; $\sum_{n=-\infty}^{+\infty} \delta(t-nT) \xrightarrow{\mathcal{F}} \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - \frac{2\pi k}{T})$
6. $a^n u[n], |a| < 1 \xrightarrow{\mathcal{F}} \frac{1}{1-ae^{-j\omega}}$ $(n+1)a^n u[n], |a| < 1 \xrightarrow{\mathcal{F}} \frac{1}{(1-ae^{-j\omega})^2}$
7. $\frac{\sin Wn}{\pi n} (0 < W < \pi) \xrightarrow{\mathcal{F}} X(\omega) = \begin{cases} 1, & 0 \leq |\omega| \leq W \\ 0, & W < |\omega| \leq \pi \end{cases}$